COUNTRY PROFILES

# Nuclear Power in Brazil

UPDATED FRIDAY, 14 NOVEMBER 2025

- Brazil has two nuclear reactors generating about 2% of its electricity.
- Its first commercial nuclear power reactor began operating in 1982.
- Construction of the country's third nuclear power reactor recommenced in November 2022 after being stalled for over seven years. In April 2023 following orders from the municipal government of Angra dos Reis, construction was once again suspended.

Privacy - Terms

2

OPERABLE REACTORS

1,884 MWe

1

REACTORS UNDER CONSTRUCTION

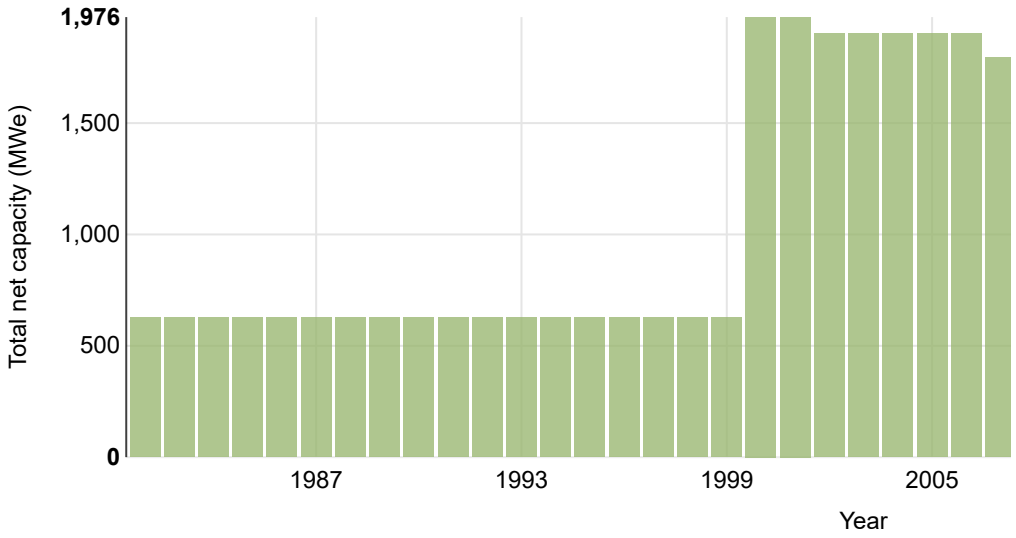
1,340 MWe

0

REACTORS SHUTDOWN

0 MWe

## Operable nuclear power capacity




# Electricity sector

**Total generation (in 2024):** 751 TWh

**Generation mix:** hydropower 422 TWh (56%), wind 108 TWh (14%), solar 70.7 TWh (9%), biofuels & waste 63.0 TWh (8%), natural gas 47.8 TWh (6%), nuclear 15.8 TWh (2%), coal 14.8 TWh (2%), oil 9.4 TWh (1%).

**Import/export balance:** 11.6 TWh net import (14.1 TWh imports, 2.5 TWh exports)

**Total consumption:** 613 TWh

**Per capita consumption:** c. 2900 kWh in 2024

*Source: International Energy Agency, The World Bank. Data for year 2024.*

Per capita electricity consumption in Brazil has grown strongly from under 1500 kWh/yr in 1990 to around 2900 kWh/yr in 2024.

The high dependence on hydro gives rise to some climatic vulnerability which is driving policy to diminish dependence on it. Major droughts in 2001 and 2015 resulted in acute shortages of power and the limiting of urban water supply. In February 2010 the government approved $9.3 billion investment in the 11.2 GWe Belo Monte hydro scheme, which opened in 2016. Its construction involved the flooding of 500 $km^2$ of the Amazon basin. The scope for further hydroelectric development is perceived to be limited.

About 40% of Brazil's electricity is produced by the national Eletrobrás Systemª. About 20% of electricity is from state-owned utilities, and the rest is from privately-owned companies.

# Nuclear power plants

### Operable reactors in Brazil

| Reactor Name | Model | Reactor Type | Net Capacity (MWe) | Construction Start | First Grid Connection |
|---|---|---|---|---|---|
| Angra 1 | 2-loop WE | PWR | 609 | 1971-05 | 1982-04 |
| Angra 2 | PRE KONVOI | PWR | 1,275 | 1976-01 | 2000-07 |

## Angra 1

Angra 1 suffered continuing problems with its steam supply system and was shut down for some time during its first few years. Its lifetime load factor over the first 15 years was only 25%, but since 1999 it has been much better. Local content was about 8%.

A request for life extension of Angra 1 was submitted in 2019. In October 2020 Westinghouse signed a contract with Eletronuclear to conduct engineering analyses to support the long-term operation (LTO) programme. Earlier in 2018, Eletronuclear requested an IAEA Safety Aspects of Long Term Operation (SALTO) review of Angra 1. The IAEA concluded that Electronuclear was “preparing unit 1 for safe LTO in a timely manner” and encouraged the operator to implement a comprehensive equipment qualification programme and a long-term workforce plan for the LTO period.

The life extension was authorized by the National Nuclear Energy Commission (CNEC) in November 2024, allowing Angra 1 to operate to 2044. CNEC said that upgrades will be made to control

systems, physical protection structures and radioactive waste management protocols. Eletronuclear said it will be investing BRL3.2 billion (USD550 million) between 2023 and 2027.

## Angra 2

Civil works on Angra 2 started in 1976, but due to a lack of financial resources and a lower than expected growth in demand, the unit only commenced operation at the end of 2000. Local content was about 40%.

In January 2025 Eletronuclear and the Brazilian Society of Nuclear Medicine signed an agreement aimed at producing lutetium-177 at Angra 2.

## Angra 3

Angra 3 was designed to be a twin of unit 2. Work started on the project in 1984 but was suspended in 1986 before full construction began. Full construction began in 2010 but was suspended in 2015 due to a corruption probe and funding issues (see below). Work recommenced in November 2022.

In November 2006 the government announced plans to complete Angra 3 and also build four further 1000 MWe nuclear plants from 2015 at a single site. Angra 3 construction approval was confirmed by Brazil's National Energy Policy Council in June 2007 and received presidential approval in July. Environmental approval was granted in March and all other approvals by July 2009. In December 2008, Eletronuclear signed an industrial cooperation agreement with Areva (now Framatome). Areva was then expected to complete Angra 3 and be considered for supplying further reactors. Areva also signed a services contract for Angra 1.

First concrete for Angra 3 was in June 2010, closely following the construction licence from the National Nuclear Energy Commission (*Comissão Nacional de Energia Nuclear*, CNEN). The plant was expected in operation at the end of 2015 after 66 months. In November 2013, in line with the 2008 agreement, it awarded a €1.25 billion ($1.7 billion) contract to Areva (now Framatome) for engineering services and components, digital instrumentation and control system, supervision of installation works and commissioning of the unit. Two Brazilian consortia bid for installation contracts. One was for electro-mechanical assembly associated with the reactor's primary system, valued at around BRL 1.31 billion ($640

million), and another was for secondary-side work, estimated at BRL 1.67 billion ($816 million). Both were awarded in February 2014.

Following a corruption probe in mid-2015, Eletrobras suspended both contracts. In mid-2016 the corruption investigations involved Eletronuclear, and then funding ran out, halting the work. In January 2017 Eletronuclear formally annulled the electro-mechanical contract, having rejected appeals by companies Andrade Gutierrez, Camargo Correa, Queiroz Galvão, UTC, Techint, Odebrecht and EBE. The unit was then about 65% complete.

In March 2017 the government announced that it planned to sell Angra 3 by 2018. The deputy energy minister said that Russian and Chinese investors had expressed interest, though Eletronuclear would be the operator. In July 2017 China National Nuclear Corporation (CNNC) was reported to be interested, along with Rosatom, Kepco and a Mitsubishi-Areva consortium. In September 2017 an agreement with CNNC was signed to promote the construction of Angra 3 and future projects. This was followed up by the signing of similar agreements with Rosatom in November 2017 and EDF in June 2018.

In November 2018 it was reported that Eletronuclear was negotiating with CNNC and State Power Investment Corp (SPIC), KEPCO, Rosatom, and a consortium comprising EDF with Mitsubishi Heavy Industries to complete the unit. A government estimate of BRL 15 billion ($3.84 billion) cost was quoted. In October 2019 Eletronuclear announced that it had shortlisted CNNC, Rosatom and EDF as possible investors. In December 2019 Nuclebrás Equipamentos Pesados shipped the final condenser for the unit.

Despite international involvement in the tender (see above), in July 2021 Eletronuclear announced that a Brazilian consortium (Agis Consortium) comprising Ferreira Guedes, Matricial and ADtranz had submitted the winning bid to complete the unit. The contract with Eletronuclear was signed in February 2022. At the end of May 2022, Eletronuclear signed an $84 million agreement with Nuclebrás Equipamentos Pesados (NUCLEP) for the manufacture and delivery of new heat exchangers for Angra 3.

Construction of the unit recommenced in November 2022.

Earlier in December 2010 Brazilian national development bank BNDES approved BRL 6.1 billion ($3.6 billion) in financing for Angra 3, covering almost 60% of the then estimated cost of BRL 9.9 billion. In December 2012 state-owned Caixa Economica Federal bank agreed to lend BRL 3.8 billion ($1.86 billion) to Eletrobras for

completion of the unit. The total estimated cost was then $7.59 billion. In June 2021 BNDES hired Angra Eurobras NES – a consortium including the Belgian engineering firm Tractebel – to complete the project.

In February 2023 the 'United Parliamentary Front for Nuclear Technology and Activities' was launched with the aim to speed up the construction on Angra 3, build another reactor at the Angra site, and build additional SMRs in the country.

In April 2023 work on Angra 3 was once again stopped following orders from the municipal government of Angra dos Reis. The mayor said he believed Eletronuclear was executing a project that was not in accordance with what the municipality approved. Eletronuclear had reportedly not paid the agreed BRL264 million (about $53 million) in socio-environmental compensation. In May 2023 Eletronuclear announced that it was committed to reversing the embargo through constructive dialogue with the Angra dos Reis City Hall.

In June 2024 it was announced that Eletronuclear had won its appeal in the Rio de Janeiro Court of Justice, removing the embargo on work at Angra 3.

In September 2024 Eletronuclear said a study by Brazil's National Bank for Economic and Social Development had concluded that the cost of abandoning construction of the part-built Angra 3 nuclear power unit could be about BRL21 billion ($3.7 billion). By contrast Eletronuclear estimated the cost of completing the project at around BRL23 billion, of which BRL12 billion had been invested.

In February 2025 the National Energy Policy Council (CNPE) postponed its decision on resuming construction of unit 3. Also in February, Eletronuclear announced a restructuring plan aimed at improving governance and the economic viability of its nuclear projects, and pushed back its target commissioning date for Angra 3 to 2031.

### Brazilian power reactors under construction

| Reactor | Model | Gross capacity | Construction start | Commercial operation |
|---|---|---|---|---|
| Angra 3 | PWR | 1405 MWe (1340 MWe net) | June 2010 (suspended)* | 2028 |

* Construction suspended between 2015 and November 2022 and suspended again from April 2023.

### Further plants

Eletronuclear has proposed building two new nuclear plants in the northeast and two more near Angra in the southeast.[2] At the end of 2009, it commenced initial siting studies. Early in 2013 two sites were under appraisal: one in the northeast on a large dam on the Sao Francisco River between Pernambuco and Bahia states for up to 6600 MWe, and one in the north of Minas Gerais state in the southeast of the country, inland from Angra, for 4000-6000 MWe. However, funding is an issue, and in May 2012 the government said that construction of any new plants would not commence until after 2020.

In January 2016 Eletronuclear took representatives from CNNC to Sergipe on the northeast coast, north of Bahia state's coastline, to look at a potential site. Rosatom has offered to consider a build-own-operate (BOO) project, as in Turkey, and in July 2014 Rusatom Overseas and Camargo Correa signed an agreement for construction of a number of facilities on the site of the existing Angra nuclear power plant, and possibly cooperating in building nuclear power units on new sites. In June 2015 Westinghouse signed an agreement with National Nuclear Energy Commission (CNEN) subsidiary, NUCLEP, for collaboration on fabrication of AP1000 reactor components in Brazil. South Korea's KEPCO is offering its APR1400.

In January 2022 the Ministry of Mines and Energy announced it was working with Eletrobras Cepel* on a siting study for new plants.

* The organization's name is the Portuguese acronym for Centre for Electric Energy Research.

In May 2025 the Minister of Mines and Energy announced plans to develop small modular reactors in cooperation with Russia's Rosatom.

## Nuclear industry development

Brazil began developing nuclear technology in 1951 under the newly-established National Research Council, but accelerated this under a military regime from 1964 to 1985. In 1970, the government decided to seek bids for an initial nuclear plant. The turnkey contract for Angra 1 was awarded to Westinghouse, and construction started in 1971 at a coastal site between Rio de Janeiro and São Paulo. This is now the Central Nuclear Almirante

Álvaro Alberto (CNAAA) power plant complex in Rio de Janeiro state, 130 km west of Rio.

In 1975, the government adopted a policy to become fully self-sufficient in nuclear technology and signed an agreement with West Germany for the supply of eight 1300 MWe nuclear units over 15 years. The first two (Angra 2&3) were to be built immediately, with equipment from Kraftwerk Union (KWU)[b]. The rest were to have 90% Brazilian content under the technology transfer agreement. To effect this, a state-owned company *Empresas Nucleares Brasileiras S.A.* (Nuclebrás) was set up with a number of subsidiaries focused on particular aspects of engineering and the nuclear fuel cycle.

However, Brazil's economic problems meant that construction of the first two Brazilian-German reactors was interrupted, and the whole programme was reorganized at the end of the 1980s. In 1988, a new company, *Indústrias Nucleares do Brasil S.A.* (INB) took over the front end fuel cycle subsidiaries of Nuclebrás. Responsibility for construction of Angra 2&3 was transferred to the utility *Furnas Centrais Elétricas S.A.* (Furnas), a subsidiary of Eletrobrás. However, Nuclen, a former Nuclebrás subsidiary that also had KWU participation, remained as the nuclear plant architect and engineering company. Construction of Angra 2 resumed in 1995, with $1.3 billion of new investment provided by German banks, Furnas and Eletrobrás. Then in 1997, the nuclear operations of Furnas merged with Nuclen to form *Eletrobrás Termonuclear S.A.* (Eletronuclear), a new subsidiary of Eletrobrás[c] and responsible for all construction and operation of nuclear power plants. After review of the policy from 2013, in May 2015 the government said that Angra 3 would be the last nuclear power plant built as a public works project, opening the way for private equity in the next four units.

Heavy equipment manufacturing remains the responsibility of former Nuclebrás subsidiary *Nuclebrás Equipamentos Pesados S.A.* (Nuclear Heavy Equipment, NUCLEP). Both NUCLEP and INB are subsidiaries of – but administratively independent of – the National Nuclear Energy Commission (*Comissão Nacional de Energia Nuclear*, CNEN), and report directly to the Ministry of Science and Technology (*Ministério da Ciência e Tecnologia*). Eletrobrás, which owns Eletronuclear, comes under the Ministry of Mines and Energy.

There is a continuing military influence on Brazil’s nuclear programme. Brazil is the only non-nuclear-weapons state in which the military leases uranium enrichment technology to the civilian

nuclear programme, and the navy drives technological advances in the nuclear field. Also Brazil is the only non-nuclear-weapons state developing a nuclear-powered submarine.

# Uranium resources & fuel cycle

Resulting from active exploration in the 1970s and 1980s, Brazil has reasonably assured resources of 210,000 tonnes[d] of uranium. There has been little investment in exploration since the mid-1980s. Three main deposits are: Pocos de Caldas (Minas Gerais state; mine closed in 1997); Lagoa Real or Caetité (Bahia state; operating since 1999); and Itataia, now called Santa Quitéria (Ceará state; phosphate as co-product; production start planned).

In the 1988 constitution the federal government reserved a monopoly over uranium resources and their development. Amendments are proposed to open up uranium exploration and mining to private enterprise, as was done in the oil and gas sector in 1995.

In December 2022 Indústrias Nucleares do Brasil (INB) signed a contract with Rosatom for the supply of 330 tonnes of uranium in the form of natural $UF_6$ to the Angra plant from 2023 to 2027. In May 2023, three contracts were signed with Westinghouse to cover the supply of advanced 16x16 fuel assemblies for Angra 1 reloads and to develop a project for the protective grid component of the fuel assembly, as well as INB supplying staff to move fuel in the USA.

## Mining

Uranium has been mined since 1982, but the only operating mine is INB's Lagoa Real/Caetité on the Cachoeira metasomatite deposit, with 340 tU/yr capacity. In 2013 all of the 192 tU production came from Caetite, where open pit operation finished in 2012 and underground operations face licensing difficulties. Modest production from heap leaching continued until 2016. It has known resources of 10,000 tU at 0.3%U.

INB commenced developing the adjacent Engenho mine in January 2017, a 200-300 tU/yr open pit operation. Production was initially planned from October 2017, but did not commence.

In January 2020, the country's energy minister reported that investment in INB would allow it to produce 150 t/yr U from Caetite,

starting in 2020, and expanding to 360 t/yr by 2023. In partnership with private companies the government aims to realise production of 1600 t/yr by 2024 from mines in Santa Quiteria. In 2020 Brazil produced 15 tU.

In 2008, INB entered an arrangement with fertilizer producer Galvani to recover uranium from phosphate mined at Itataia/Santa Quitéria[3] in the north of the country. The open pit mine was expected to produce 970 tU per year from 2016, and ramp up to 1270 tU/yr as by-product or co-product of phosphate, but licensing has been slow. Reserves are 79,000 tU at 0.1%U, with 140,000 tU resources quoted elsewhere. In March 2022 the Santa Quiteria project was accepted for environmental review by the Brazilian Institute of Environment and Renewable Natural Resources (IBAMA). INB expected the Itataia facility to begin operation in 2023, but it is yet to receive a licence from IBAMA.

All mined uranium is used domestically, after conversion and most enrichment abroad.

## Conversion

Apart from the navy's small experimental plant at Aramar, all conversion is by Areva in France. The $UF_6$ has mostly been sent to Urenco for enrichment, but increasingly can be enriched at Brazil's Resende plant.

## Enrichment

Most enrichment has been undertaken by Urenco in Europe or USA.

In the early 1980s, the Brazilian Navy started a nuclear propulsion program and undertook the development of centrifuge enrichment to 1989. A demonstration plant was built at the Aramar Experimental Center in Iperó (São Paulo state), which remains a naval facility to provide fuel enriched to less than 20% for the submarine program. Enrichment here is reported to be to 5% U-235.

Using the enrichment technology developed by the navy at Aramar, and with centrifuges built by the navy and leased to INB, an industrial enrichment plant at Resende[e] is intended to cater for much of the needs of the Angra reactors. The centrifuges are domestically-developed and similar to Urenco technology.

Construction on stage 1 of Resende officially commenced in 2006 by INB. This stage of the plant was originally expected to consist of four modules totalling 115,000 SWU/yr, with each module made up of four or five cascades of 5000-6000 SWU/yr. Production commenced in 2009[4] and it produced 730 kg of 4% enriched uranium in that year. In 2012 three cascades were operating; a seventh was commissioned in August 2018, and INB then said that the first stage would involve a total of 10. Once completed phase 1 will provide 80% of the fuel requirements for Angra 1.

A further 30 cascades are planned as stage 2. So far, BRL 560 million (approx. $135 million) has been invested. The total investment cost for both stages is expected to be BRL 3 billion (approx $730 million).

In June 2016 INB contracted with Argentina's Conaur, a CNEA subsidiary, to export four tonnes of enriched uranium oxide for the Carem reactor.

INB also operates a reconversion plant to make $UO_2$ powder, with a capacity of 160 t/yr.

### Fuel fabrication

INB's fuel fabrication plant designed by Siemens is also at Resende, with capacity of 120 tonnes per year pellet production and 280 t/yr fuel assembly production.

## Radioactive waste management

The National Nuclear Energy Commission (*Comissão Nacional de Energia Nuclear*, CNEN) is responsible for management and disposal of radioactive wastes. Legislation in 2001 provides for repository site selection, construction and operation for low- and intermediate-level waste. A long-term solution for these is to be in place before Angra 3 is commissioned.

Used fuel is stored at Angra pending formulation of policy on reprocessing or direct disposal. It is not considered a waste. In April 2025 the National Nuclear Energy Commission granted Eletronuclear a 40-year licence extension for the dry storage of used fuel at the Angra plant.

Used fuel from research reactors, all LEU, is returned to the country of origin.

# Research and development

CNEN's Directorate of Research and Development (DPD) is responsible for all fuel cycle, reactor technology, radioisotopes, and related R&D. Five nuclear research centres carry out various R&D work. At the Nuclear Energy Research Institute (*Instituto de Pesquisas Energéticas e Nucleares*, IPEN), São Paulo, there are two research reactors (including a 5 MW pool type reactor) and a cyclotron, with radioisotope production.

In May 2013 Argentina's INVAP was awarded contracts to build the RA-10 research reactor in Argentina and the Brazil Multipurpose Reactor (RMB) there, with Australia's OPAL reactor being the reference design for both. The two reactors will be used for the production of medical radioisotopes, as well as irradiation tests of advanced nuclear fuel and materials, and neutron beam research. Under a related contract signed in January 2012, Brazil's Intertechne is developing the conceptual and basic design of buildings, systems and infrastructure for the RMB. Construction started in 2018 in the municipality of Iperó in São Paulo state. The research reactor project is part of the growing bilateral cooperation in nuclear energy between Argentina and Brazil.

At CTMSP (*Centro Tecnológico da Marinha em São Paulo*) – the Navy's Aramar Technology Center at São Paulo – a prototype reactor for naval propulsion is being developed by the Nuclear Power Generation Laboratory (LABGENE). A reactor pressure vessel for the prototype has been supplied by NUCLEP (*Nuclebrás Equipamentos Pesados S.A.*). At one stage it was reported that this programme was being redirected into possible applications for small power plants in the northeast of the country. Assembly of the Nuclear Propulsion Prototype Reactor began in October 2020. The reactor, as well as generators, electric motor and other systems deployed on a nuclear submarine will be tested at CTMSP. If testing is successful, a similar reactor will be installed in the *Alvaro Alberto* submarine, being developed by the Brazilian navy as part of the SN-BR project.

Brazil has been involved in the Generation IV International Forum, and in the IAEA INPRO (International Project on Innovative Nuclear

Reactors and Fuel Cycles) program, both developing new-generation reactor designs and systems. CNEN is also involved with Westinghouse in developing the IRIS modular reactor.

In June 2022, Eletronuclear renewed the 2018 memorandum of understanding with France's EDF. The agreement will see mutual collaboration on the development of nuclear energy projects in Brazil over the following five years.

In March 2025, CNEN initiated a three-year project to demonstrate the feasibility of a 3–5 MW microreactor, intended for remote operation over 10 years without refueling. The Institute of Energy and Nuclear Research (IPEN) will lead research on materials, whilst the Institute of Nuclear Engineering (IEN) will lead on assembling a subcritical test unit. Various universities and the Brazilian Navy and Amazônia Azul Defense Technologies are also involved in the project.

In March 2025 the National Nuclear Energy Commission launched a three-year project to develop a 3-5 MW containerized microreactor, aiming for national design autonomy and potential use in remote regions, with research led by CNEN, IPEN, and IEN.

## Regulation, safety and non-proliferation

The main legislation is the National Policy on Nuclear Energy, 1962, which established state control over nuclear materials. The National Nuclear Energy Commission (CNEN) was set up in 1956 and reported initially to the Presidential Secretary for Strategic Affairs but now comes under the Ministry of Science and Technology. In 1974, legislation was passed that established CNEN as the country's nuclear regulator. CNEN's Directorate of Radiation Protection and Safety (DRS) is responsible for licensing and supervision of all nuclear facilities.

In May 2021 the Brazilian government announced the creation of a new regulator, separate from CNEN. Once established, the National Nuclear Safety Authority (*Autoridade Nacional de Segurança Nuclear*, ANSN) will have administrative, technical and financial autonomy from CNEN and will be charged with setting requirements relating to safety, radiation protection and security for all nuclear facilities and activities. CNEN will remain in charge of planning and overall policy and advocacy for nuclear energy. However, as of June 2025 ANSN had not been established.

In 1989, the Brazilian Institute of Environment and Renewable Natural Resources (*Instituto Brasileiro do Meio Ambiente e dos Recursos Naturais Renováveis*, IBAMA) was created to carry out environmental licensing of all facilities (including nuclear), but CNEN remains a co-authority on radiation aspects of nuclear licensing. IBAMA comes under the Ministry of the Environment.

The National Council for Scientific and Technological Development (*Conselho Nacional de Desenvolvimento Científico e Tecnológico*, CNPq) was set up in 1951 as the National Research Council to develop nuclear technology in Brazil.

In January 2025 Brazil approved funding for the Nuclear Physical Security Training Centre at the Institute of Nuclear Engineering, designed to train specialists in nuclear facility protection and transport security, with international cooperation and a three-year completion target.

## Non-proliferation

Brazil is a party to the Nuclear Non-Proliferation Treaty (NPT) since 1998 as a non-nuclear-weapon state (it signed in 1995), but has been a party to the Tlatelolco Treaty since 1967[f]. Its reservations on the NPT were that it did not exempt peaceful nuclear explosions for civil engineering, and it was weak on global disarmament. Nevertheless it had a safeguards agreement with the International Atomic Energy Agency (IAEA) since 1968 (INFCIRC 110 & 147).

While the Tlatelolco Treaty did not include Argentina, the relationship in that direction was always rivalry rather than any arms race. With the advent of civilian government in 1985 Brazil and Argentina signed a Joint Declaration on Nuclear Policy to address “the growing difficulties arising in the international supply of nuclear equipment and materials.” In the course of pursuing independence in its nuclear fuel cycle through the 1970s and 80s Brazil had firmly denied aspirations for nuclear weapons. Nevertheless the USA was active in attempting to counter the country’s agreement with West Germany because it included the possibility of supplying enrichment technology.

Following a new constitution in 1988, the country again renounced development of nuclear weapons[g] and, in 1991, the Brazilian-Argentine Agency for Accounting and Control of Nuclear Materials (ABACC) was set up. This led to the 1991 Quadripartite Agreement (INFCIRC 435) among Brazil, Argentina, ABACC and the IAEA

which entered force in 1994 with full-scope safeguards under IAEA auspices including naval facilities.

In 1996, Brazil became a member of the Nuclear Suppliers Group.

Brazil has not accepted the Additional Protocol in relation to its safeguards agreements with the IAEA on the principled basis that the international system promotes non-proliferation rather than the more fundamental question of nuclear disarmament, and practically because its application to the submarine program is unclear and because it could cut across ABACC (Argentina also has not signed the Additional Protocol).

In 2010 Brazil, with Turkey, signed an agreement with Iran – the Teheran Declaration – to swap its 20% enriched uranium for foreign fuel for the Teheran research reactor, alleviating concerns about Iran's intentions in enriching uranium to that level. This was not acted upon, but led to later agreements with the five permanent members of the UN Security Council plus Germany (P5+1).

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# Notes & references

## Notes

a. Eletrobrás (Centrais Elétricas Brasileiras S.A.) was set up in 1962 as a holding company controlled by the Ministry of Mines and Energy. The federal government currently holds a 54% shareholding in the company. The Eletrobrás System companies are responsible for over 39,400 MWe of the installed capacity in the country, which represents 38% of the total Brazilian capacity. It holds 50% of the shares in Itaipu Binational, which runs the 14 GWe Itaipu hydroelectric dam. Eletrobrás is also the owner of Eletronuclear (*Eletrobrás Termonuclear*), the Brazilian nuclear utility. [Back]

b. In 1969, Siemens and AEG merged their nuclear activities, forming Kraftwerk Union (KWU). In 1977 AEG sold all its shares in KWU to Siemens. In 1987, Siemens-KWU was integrated into Siemens' Power Generation Group and, in 2001, Siemens merged its nuclear activities with Framatome to form Framatome ANP, which was later rebranded as Areva NP. In 2009, Siemens announced its intention to sell its 34% interest in the joint venture to Areva. [Back]

c. When Eletronuclear was formed (from Nuclen and Furnas), Siemens sold its 25% stake in Nuclen to Eletrobrás. [Back]

d. Reasonably Assured Resources, to US$ 130/kg. [Back]

e. The enrichment plant is part of the Resende Nuclear Fuel Factory (*Fábrica de Combustível Nuclear*, FCN) or INB Resende. [Back]

f. The 1967 Treaty for the Prohibition of Nuclear Weapons in Latin America and the Caribbean, known as the Treaty of Tlatelolco, was signed by all Latin American countries other than Argentina and Cuba in 1967. The Treaty came into force in April 1969. [Back]

g. In the 1970s, Brazil's military government pursued a covert nuclear weapons program. This program was ended by the civilian government, which came to power in 1985 under President José Sarney. [Back]

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# Related information

Uranium Production by Country